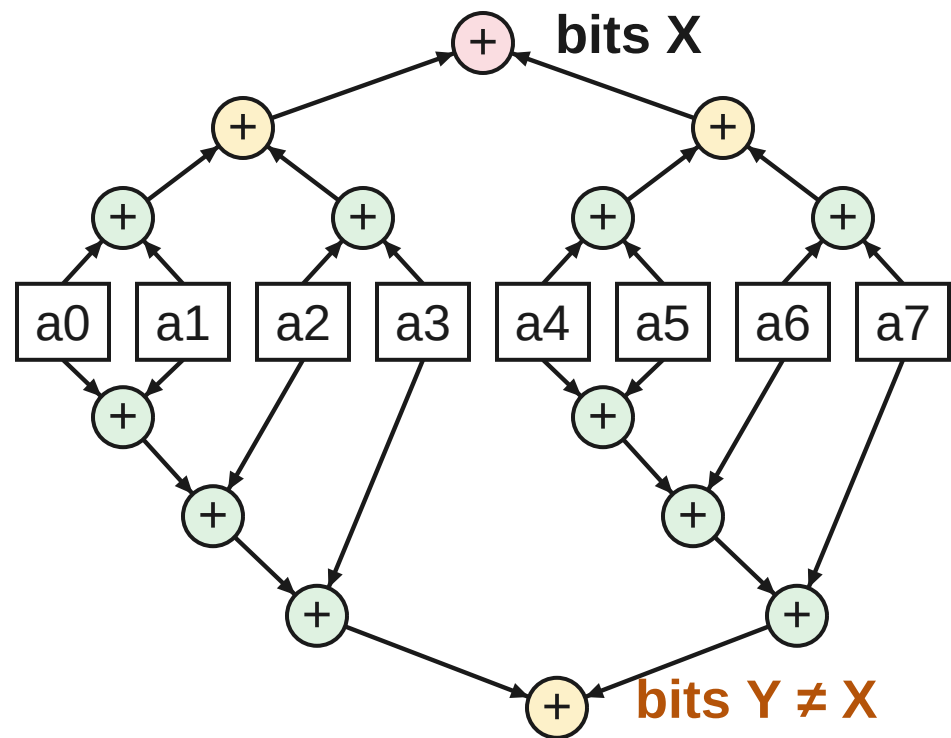


layout A — 4 threads $\times$ 2 elements, 2 warps

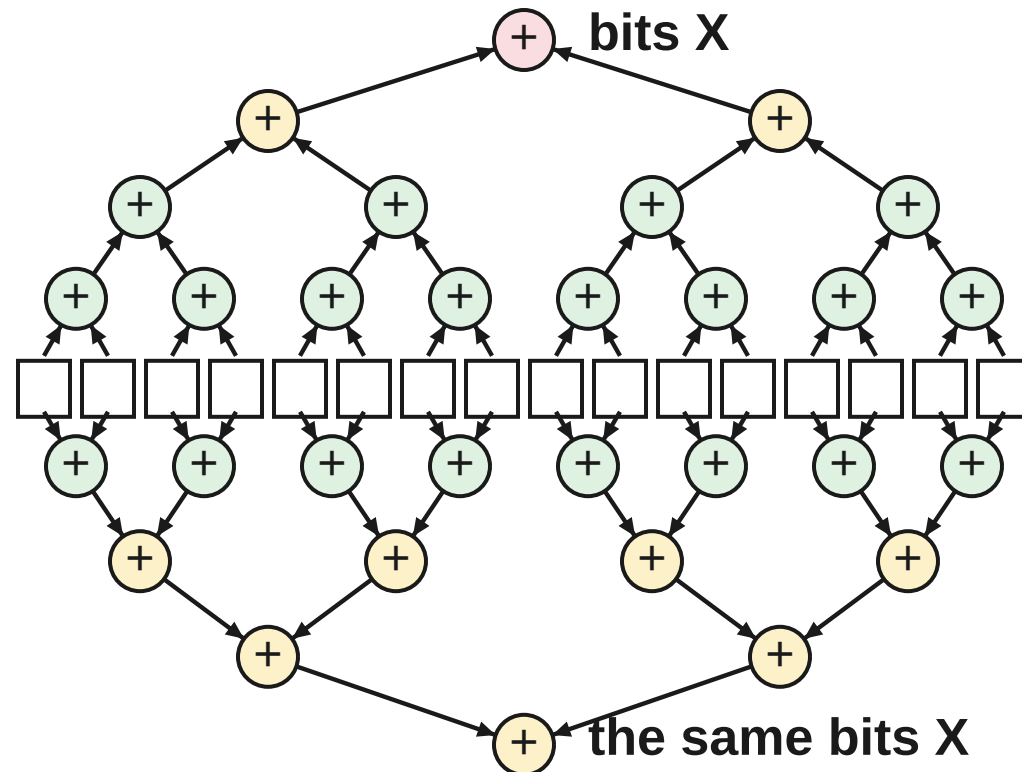


layout B — 2 threads $\times$ 4 elements, 1 warp

○ within a thread (registers) ○ within a warp (lane shuffle) ○ across warps (shared memory)

(a) Order free: two layouts, two trees

layout A — 4 threads $\times$ 4 elements, 2 warps

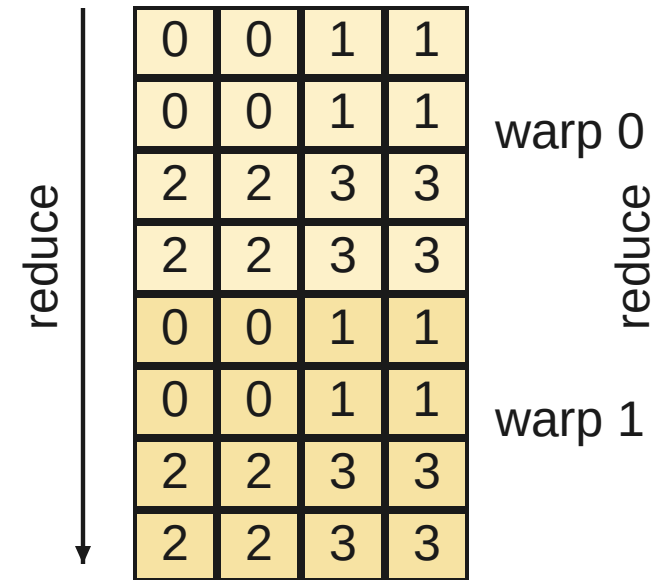


layout B — 8 threads $\times$ 2 elements, 1 warp

○ within a thread (registers) ○ within a warp (lane shuffle) ○ across warps (shared memory)

(b) Order pinned: two layouts, equivalent tree

before



after



(c) The layout optimization